

Smart City Bike Sharing Analysis Report

Executive Summary:

This report presents a descriptive and exploratory analysis of a smart city bike-sharing dataset using Power BI. The objective of the analysis was to evaluate network-scale station performance, compare operational efficiency across contracts/networks, identify service risk patterns such as empty or full stations, and derive insights that can support better bike redistribution and station management. The dashboard reveals that the bike-sharing network is largely operational, with a high share of open stations and a sizable installed capacity across 25 contracts/networks. However, bike availability is uneven across locations, and several stations remain either empty, full, or operationally imbalanced. Contract-level comparisons show that some contracts maintain stronger station balance, while others appear to require better redistribution and monitoring. Strategic recommendations are provided to improve service balance, reduce empty/full station risk, and strengthen operational consistency across contracts/networks.

Business Objective:

- To analyze overall bike-sharing network performance
- To compare station availability and operational health across contracts/networks
- To identify high-risk stations with empty, full, or closed conditions
- To evaluate service balance using station utilization patterns
- To derive business insights for redistribution and operational improvement

Dataset Name: Smart City Bike Sharing Data

Data Volume Overview:

- Total Contracts/Networks analyzed: **25**
- Total installed bike stand capacity: **approximately 57K**
- Open station rate: **approximately 92%**
- Bike availability rate: **approximately 34%**

Analysis Type: Descriptive and Exploratory Analysis

Tools Used:

- Microsoft Power BI
- Power Query
- DAX
- Data Modelling using Star Schema

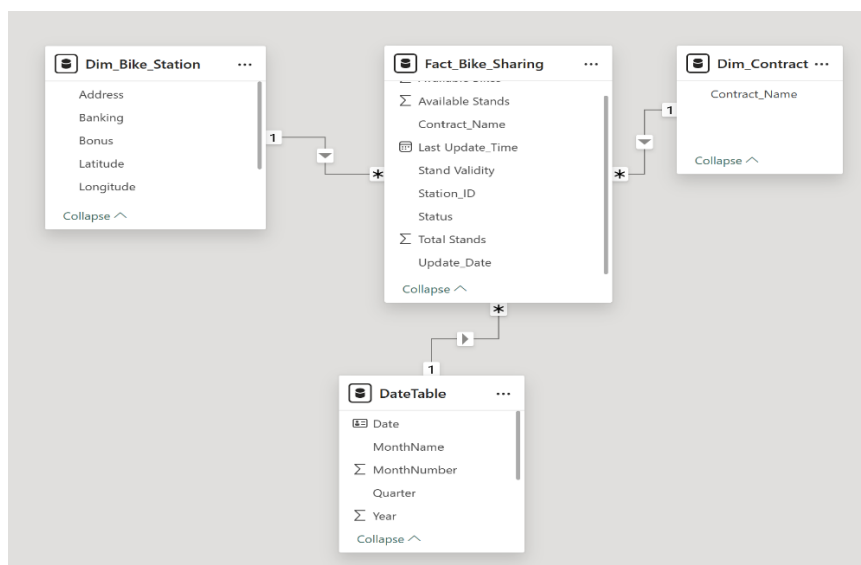
Data Cleaning and Preparation Summary:

- Missing and inconsistent values handled during Power Query transformation
- Column names and data types standardized for analysis
- Fact and dimension tables created for bike-sharing operations, station attributes, contracts, and date analysis
- Relationships defined using a star schema structure

Data Modelling:

The model was structured around a central fact table containing operational station metrics such as available bikes, available stands, total capacity, status, and update date. Supporting dimension tables were created for station attributes, network/contract information, and calendar-based analysis.

Figure 1: Data Modelling



DAX Measures / Calculated Columns:

DAX Measure / Column Name	Type	Purpose
Total Contracts/Networks	Measure	Calculates the total number of unique contracts/networks in the dataset. Used in KPI cards.
Total Stations	Measure	Calculates the total number of unique stations. Used to represent network size.
Total Valid Records	Measure	Counts all cleaned and valid records included in the final analytical dataset.
Total Capacity	Measure	Calculates the total installed bike stand capacity across all stations. Used in KPI cards.
Available Bikes	Measure	Calculates the total number of currently available bikes across the network.
Available Stands	Measure	Calculates the total number of currently free stands across the network.
Open Stations	Measure	Counts the number of stations currently marked as OPEN.
Closed Stations	Measure	Counts the number of stations currently marked as CLOSED.
Open Station Rate %	Measure	Calculates the percentage of stations that are operational/open. Used in KPI cards.
Bike Availability Rate %	Measure	Calculates the proportion of available bikes relative to total station capacity. Used in KPI cards.
Balanced Stations	Measure	Counts stations that fall within the balanced utilization range.
Balanced Station Rate %	Measure	Calculates the percentage of stations considered operationally balanced. Used in KPI cards.

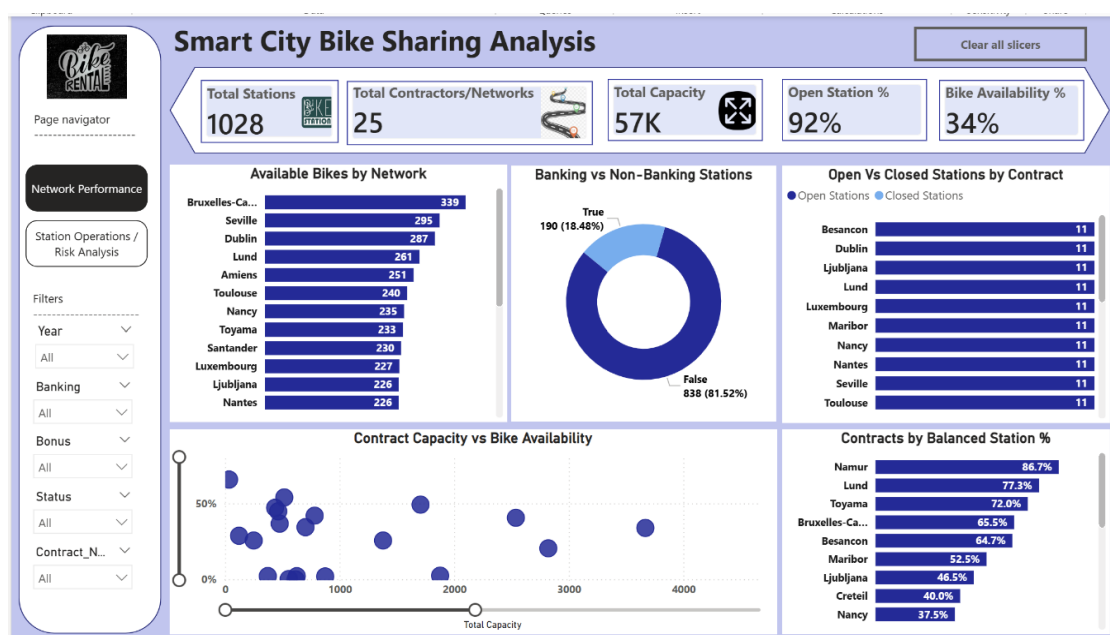
Empty Stations	Measure	Counts stations with zero available bikes. Used in risk analysis.
Full Stations	Measure	Counts stations with zero available stands. Used in risk analysis.
Latest Update	Measure	Returns the latest available timestamp in the dataset. Used to show data freshness.
Utilization %	Calculated Column	Measures the ratio of available bikes to total station capacity for each station. Used in utilization analysis.
Utilization Band	Calculated Column	Classifies stations into ranges such as 0–10%, 11–30%, 31–70%, and 71–90% for distribution analysis.
Service Risk	Calculated Column	Categorizes stations into operational groups such as Empty, Full, Balanced, or Closed. Used in the high-risk station table.
Time Slot	Calculated Column	Groups records into Morning, Afternoon, Evening, and Night based on update time. Used in time-slot analysis.

KPI Metrics:

- Total Contracts/Networks: **25**
- Total Capacity: **57K**
- Open Station Rate: **92%**
- Bike Availability Rate: **34%**
- Balanced Station Rate: **47%**
- Total Stations: 1028
- Total Valid Records: 2856

Report Dashboard:

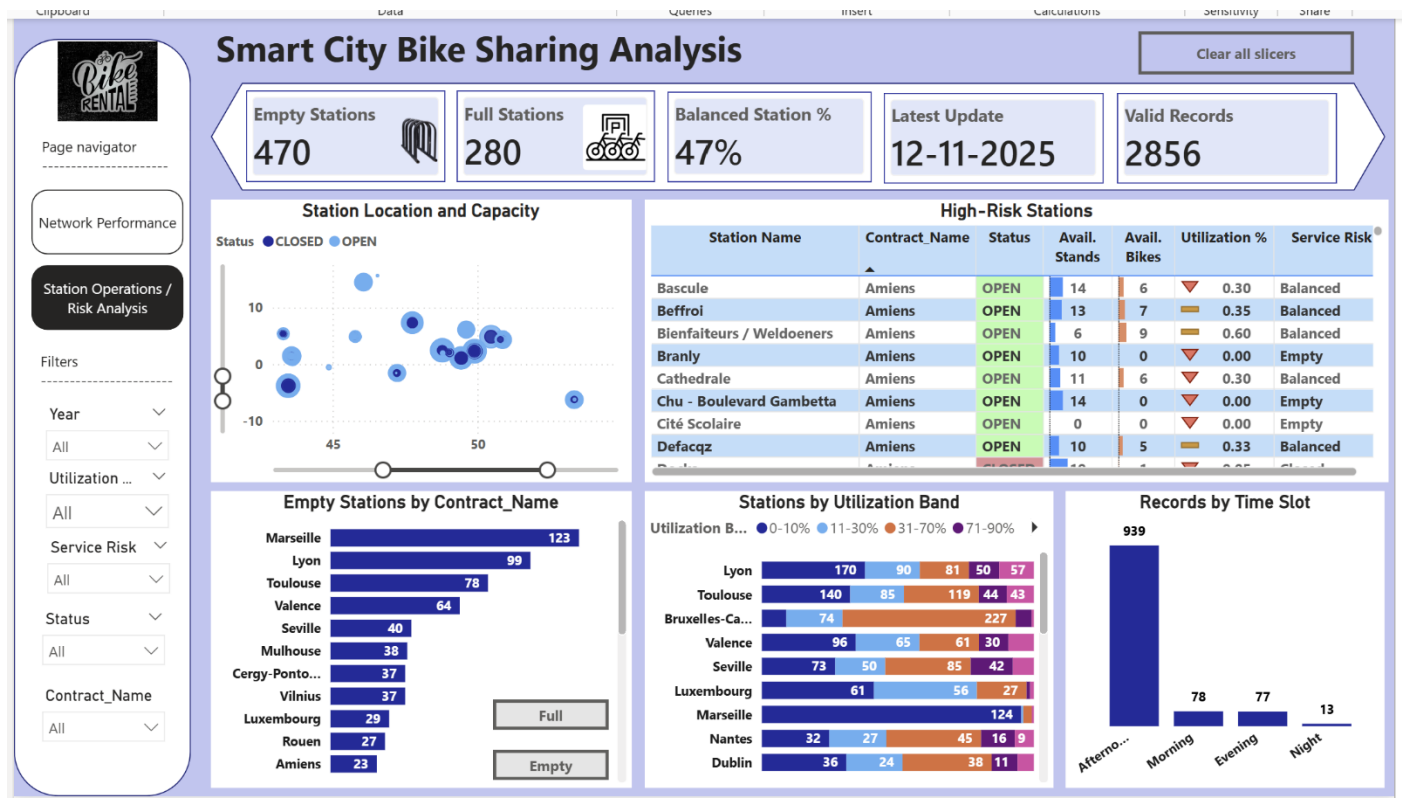
Figure 2: Network Performance Dashboard



Visual	Business Problem Addressed	Visual Analysis	Insight	Business Solution / Recommendation
1. Available Bikes by Contract/Network	Which contracts/networks currently have stronger bike availability?	The bar chart shows that Bruxelles-Capitale, Seville, Dublin, Lund, and Amiens lead in visible bike availability. These contracts currently hold more available bikes than the others shown in the chart.	• Bruxelles-Capitale leads, while Ljubljana and Nantes sit at the lower end of the visible ranking • Available-bike stock dominates in the top contracts • Service weakness is moderate in the lower group.	Benchmark lower-performing contracts against high-performing ones. Review redistribution patterns, local demand pressure, and station-level balance in weaker contracts.
2. Banking vs Non-Banking Stations	What is the distribution of station infrastructure features across Contract/Network?	The donut chart shows that non-banking stations account for 81.52% of stations, while banking-enabled stations account for 18.48% .	• Non-banking stations dominate, while banking-enabled stations remain limited • The infrastructure mix is clearly skewed and the feature gap is severe.	Review whether banking-enabled facilities should be expanded in high-demand or central stations to improve convenience and service accessibility.
3. Open vs Closed Stations by Contract/Network	Which contracts/networks are maintaining stronger station operability?	The chart compares open and closed stations across contracts. The visible top contracts show high open-station counts, with very low closed-station presence in the selected view.	• The visible top contracts/networks show strong open-station dominance • Closed stations are negligible • The open-closed gap is wide.	Contracts with weaker open/closed performance should be reviewed for maintenance issues, downtime frequency, and operational reliability.
4. Contract/Network's Capacity vs Bike Availability	Does higher installed capacity lead to better bike availability?	The scatter chart compares total contract/network capacity with bike availability %. High-capacity contracts/networks are mostly clustered around moderate availability levels, while lower-capacity contracts/networks show more variation.	• Higher-capacity contracts/networks do not consistently dominate bike availability • Moderate availability dominates most large contracts/networks, but a few outliers show very low availability.	Evaluate contract performance using both capacity and bike availability . Large contracts/networks should also be measured on distribution efficiency and balance quality.
5. Contracts/ Networks by Balanced Station %	Which contracts/networks are operating with healthier station balance?	The bar chart shows Namur, Lund, Toyama, Bruxelles-Capitale, and Besançon as the strongest visible contracts/networks by	• Namur leads, while Nancy is lowest among the visible contracts/networks • Balanced stations dominate more strongly at the top • Imbalance is low in the leaders and	Use Balanced Station % as a core KPI for contract-level performance. Lower-ranked contracts/networks should be targeted for more frequent

		balanced station percentage.	moderate lower down.	redistribution and monitoring.
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Figure 3: Station Operations/ Risk Analysis Dashboard



Visual	Business Problem Addressed	Visual Analysis	Insight	Business Solution / Recommendation
1. Station Location and Capacity	Where are stations located, and how is capacity distributed spatially?	The location visual plots stations by geographic position and shows OPEN/CLOSED status while reflecting station size. Stations are clustered in certain coordinate zones rather than spread evenly.	- The central cluster dominates, while more isolated locations appear weaker Open stations dominate the spatial view - Closure severity is low.	Use spatial analysis to identify service clusters, under-covered zones, and high-risk operational pockets. This can support targeted field planning and station deployment.
2. High-Risk Stations Table	Which specific stations require immediate operational attention?	The table lists station name, contract/network, status, available stands, available bikes, utilization %, and service risk. Stations categorized as Empty , Balanced , or other risk groups are visible for direct review.	- In the visible Amiens sample, balanced stations dominate while a few empty stations stand out as weaker cases OPEN status dominates - Closure severity is negligible.	Use this table as a daily control view for rebalancing, inspection, and maintenance. High-risk stations should be prioritized first.
3. Full/Empty Stations by Contract_Name	Which contracts/networks have the highest	The chart shows that contracts/networks such as Marseille , Lyon ,	Marseille and Lyon dominate empty-station counts	Prioritize rebalancing operations in contracts/networks with

	concentration of problematic stations?	Toulouse, Valence, and Seville contribute strongly to the current full/empty station count.	• Service shortage severity is high at the top • The gap versus lower contracts/networks is substantial.	the highest concentration of empty/full stations. These contracts/networks need stronger intervention and closer monitoring.
4. Stations by Utilization Band	How are stations distributed across low, medium, and high utilization conditions?	The stacked bar chart classifies stations into ranges such as 0–10%, 11–30%, 31–70%, and 71–90% by contract/network. Some contracts/networks have more stations in the middle ranges, while others show heavier concentration in low or extreme bands.	• Bruxelles-Capitale shows stronger mid-band balance, while Lyon and Toulouse carry heavier low-utilization pressure • Imbalance severity is moderate to severe in low-band-heavy contracts/networks.	Track utilization-band distribution regularly. Aim to increase the share of stations in the balanced middle range and reduce extreme low/high utilization categories.
5. Records by Time Slot	What operational time pattern is visible in the dataset?	The chart shows a strong concentration of records in the Afternoon time slot, followed by much smaller counts in Morning, Evening, and Night.	• Afternoon dominates, while Night is minimal • The concentration gap is severe which means the time-slot pattern reflects update timing rather than balanced operational activity.	Treat this visual as operational context, not demand forecasting. Use it to explain when records are more strongly represented, but avoid overinterpreting it as usage behaviour.

Limitations

- The dataset is operational snapshot data, not trip-history or revenue data
- Station numbering may repeat across contracts/networks, so station counting logic must use a unique key
- Time-based insights should be treated as operational context, not long-term behavioural trend analysis

Conclusion

The smart city bike-sharing analysis demonstrates that the network is broad, mostly open, and supported by substantial installed capacity. However, the analysis also shows that service availability is uneven and that a meaningful number of stations are operationally imbalanced. Contract-level comparisons reveal stronger and weaker contracts/networks in terms of balanced station distribution, while station-level analysis highlights where direct operational action is needed. Overall, the project successfully translates raw station data into practical decision-support insights for mobility operations. The strongest improvement opportunities lie in redistribution planning, station balance monitoring, and correcting high-risk empty/full station conditions.